

Capacity Planning

Cloud Computing

2026

Reference

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- Jamsa, K. (2022). Cloud computing. Jones & Bartlett Learning.

Background of Capacity Planning

- Cloud computing promotes the idea of **on-demand and seemingly unlimited resources**.
- This “infinite resource” concept is enabled through:
 - Flexible resource pooling
 - Virtualization
 - Auto-scaling mechanisms
- In reality, cloud providers have **limited physical resources** at the data center level.
- The impression of unlimited resources is created through **strategic resource management and efficient utilization**.

Capacity Planning

- Capacity planning is a strategy to ensure that **available resources are sufficient** to meet current and future demand.
- The main goal is to provide the **right amount of computing resources** at the **lowest possible cost**.
- Application resource needs often change over time, depending on:
 - User behavior
 - Workload patterns
 - Application requirements
- In cloud computing, capacity planning applies to both:
 - Computing resources, such as CPU, memory, and storage
 - Networking resources, such as bandwidth and connectivity

Capacity Planning in Traditional Computing

- Capacity planning was usually done in isolated silos or separate data centers.
- Stakeholders had limited control over resource planning.
- Fixed resource capacity made it difficult to handle changing application demands.
- Poor planning could lead to:
 - Over-provisioning, causing resource wastage and high costs
 - Under-provisioning, causing poor performance and resource shortages

Capacity Planning in Cloud Computing

- Cloud removes the need for small isolated infrastructure silos.
- Users can self-provision resources based on their actual needs.
- Multi-tenancy and resource sharing improve efficiency and reduce waste.
- Cloud gives the impression of limitless resources, but physical resources are still limited.
- Therefore, capacity planning remains important to ensure cost efficiency, performance, and service reliability.

Capacity Planning Levels in Cloud Computing

- Capacity planning in cloud computing is performed at **two levels**.
- At the **first level**, each consumer of a cloud service does his own capacity planning.
- At the **second level**, the cloud service provider analyzes the capacity requirements of all consumers together and makes the ultimate overall capacity planning.

Consumer-Level Capacity Planning

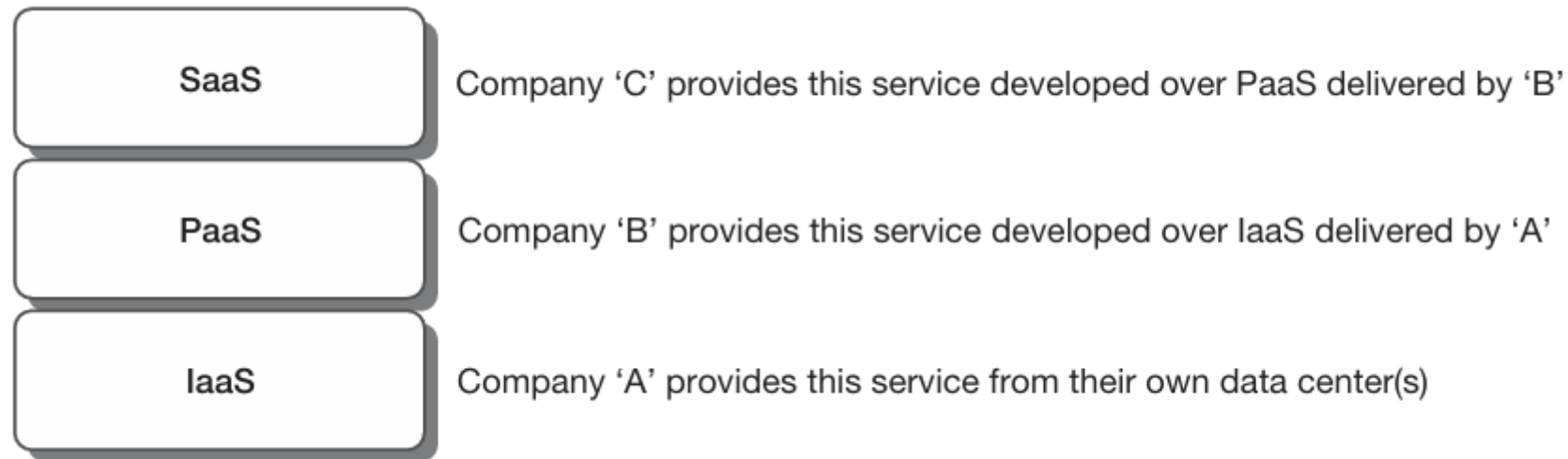
- Cloud consumers estimate their own resource needs based on application demand and budget.
- **IaaS consumers** usually plan and reserve a fixed amount of virtual resources.
- **PaaS and SaaS consumers** often use dynamically metered resources based on actual usage.
- Service requirements stated in the **SLA** help providers understand expected demand.
- Consumers do not need to plan physical infrastructure, but poor estimation can increase billing costs.

Provider-Level Capacity Planning

- The **IaaS provider** performs the actual physical capacity planning.
- The provider analyzes:
 - Resource requirements of all consumers
 - SLA agreements
 - Current and future service demands
- Infrastructure teams monitor usage continuously and plan future resource procurement.
- This allows cloud consumers to scale up or down during runtime with lower risk.

Capacity Planning Responsibilities Across Cloud Layers

- Capacity planning responsibility differs across **SaaS**, **PaaS**, and **IaaS** layers.
- Example: Consider three different companies who are providers of different cloud services under some agreement.

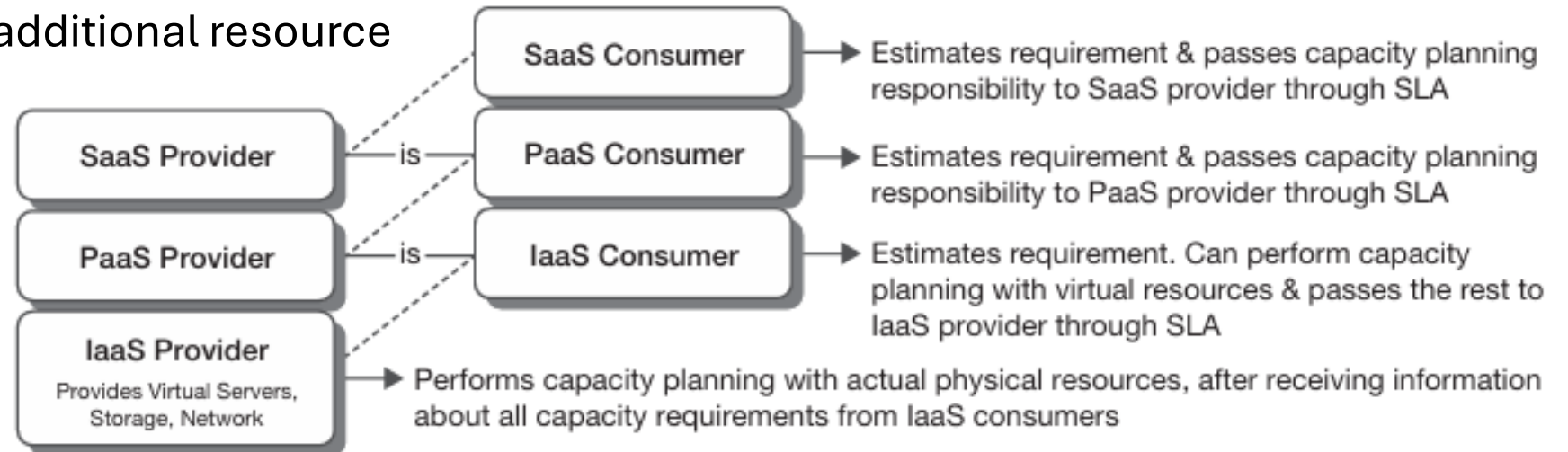


Capacity Planning Responsibilities Across Cloud Layers

- End users and upper-layer providers must communicate expected demand through **SLA agreements**.
- Each cloud layer has three possible approaches:
 - Estimate exact future demand.
 - Transfer full responsibility to the lower-layer provider.
 - Manage demand up to a certain level and rely on the lower layer for extreme cases.
- SaaS and PaaS providers should estimate future demand carefully to avoid passing excessive risk to the IaaS provider.
- The **IaaS provider is the final layer of responsibility** because it manages the actual physical resources.
- If upper layers fail to estimate demand properly, the IaaS provider may struggle to support sudden large-scale resource requests.

Role of SLA in Capacity Planning

- Service Level Agreement (SLA) defines the contract between cloud service provider and consumer.
- In cloud computing, SLA should clearly mention:
 - Expected resource demand
 - Service quality and scope
 - Limits of resource availability
 - Conditions for additional resource provisioning

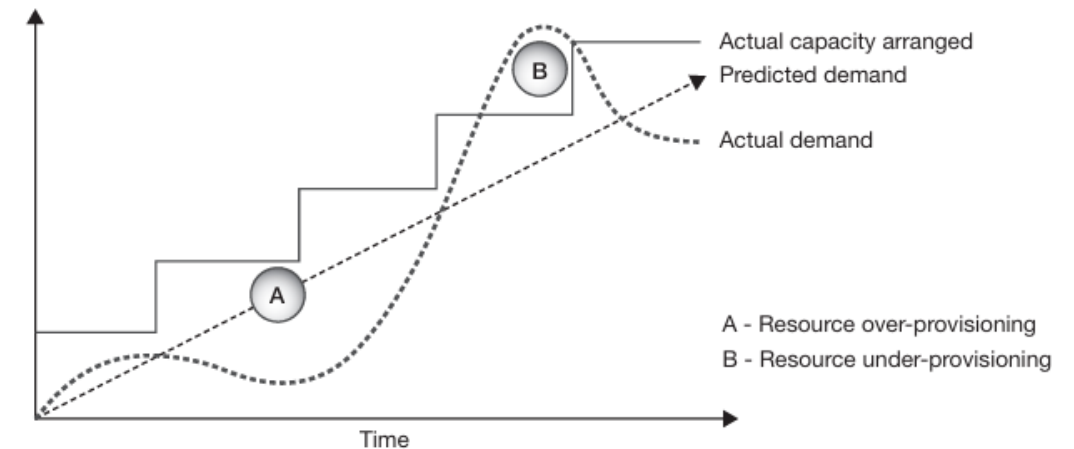


Role of SLA in Capacity Planning

- Consumers should declare both:
 - Normal or average resource requirements
 - Peak demand periods and estimated extra resources needed
- Providers should explain:
 - When extra resource supply may become difficult
 - What actions will be taken during resource shortages
- SLA helps transfer and coordinate capacity planning across layers:
 - SaaS consumer → SaaS provider
 - SaaS provider → PaaS provider
 - PaaS provider → IaaS provider
- All stakeholders should participate in SLA negotiation to define realistic service expectations.

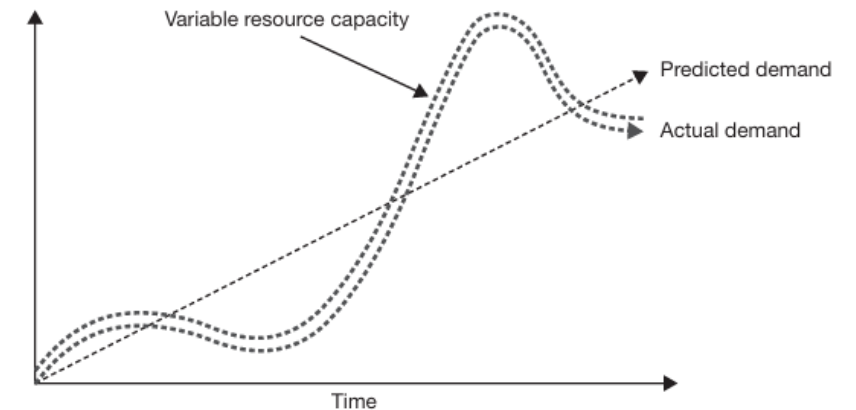
Traditional Fixed Cost Computing Model

- Traditional capacity planning used a fixed cost model.
- Organizations bought or outsourced computing infrastructure based on long-term estimated demand.
- This often created two major problems:
 - Over-provisioning: too many resources during low demand
 - Under-provisioning: insufficient resources during traffic spikes
- Over-provisioning causes idle resources and wasted investment.
- Under-provisioning causes poor performance, service unavailability, and lost business opportunities.



Modern Variable Cost Computing Model

- Cloud computing allows consumers to use a variable cost model instead of a fixed cost model.
- Computing resources are delivered as services, so consumers do not need to buy permanent infrastructure.
 - Consumers rent resources and pay based on actual usage.
 - Additional resources can be provisioned when demand increases.
 - Resources can be released when they are no longer needed.
- Cloud providers support many clients with different demand patterns. Since not all clients reach peak demand at the same time, providers can share resource pools efficiently.



Capacity Planning Approaches: Traditional vs Cloud

- Capacity planning is a critical task that requires expertise, time, and budget.
- A well-planned system improves resource utilization and reduces idle capacity.
- Better utilization leads to lower computing costs and more reliable service.
- There are two approach:
 - Traditional Approach: N+1 Rule
 - Cloud-Specific Approach

Traditional Approach: N+1 Rule

- Traditional systems often use the **N+1 rule** for capacity maintenance.
- If **N nodes** are required to handle current demand, one extra node is added as backup.
- The additional **+1 node** helps protect the system from:
 - Sudden demand spikes
 - Hardware failure
 - Service interruption
- This buffer reduces the risk of overloaded nodes and failed user requests.
- However, the N+1 rule may not always be enough for rapidly changing business demands.

Cloud-Specific Approach

- Cloud computing offers a more flexible approach through **virtual resources** and **auto-scaling**.
- Auto-scaling adjusts resource capacity based on application demand.
- Cloud consumers do not need to manually monitor and maintain capacity all the time.
- The system can automatically add or release resources when needed.
- However, IaaS providers still manage physical resources using traditional capacity maintenance principles.

Steps for Capacity Planning

- Capacity planning helps service providers manage computing resources as a utility service.
- There are three main steps in capacity planning:
- **Step 1: Determine Expected Demand**
 - Identify expected resource usage patterns.
 - Study how demand changes over time.
 - Estimate future workload and service needs.
- **Step 2: Analyze Current Response to Load**
 - Evaluate current resource capacity.
 - Observe how applications respond to normal load and overload.
 - Identify whether additional capacity is needed.

Steps for Capacity Planning

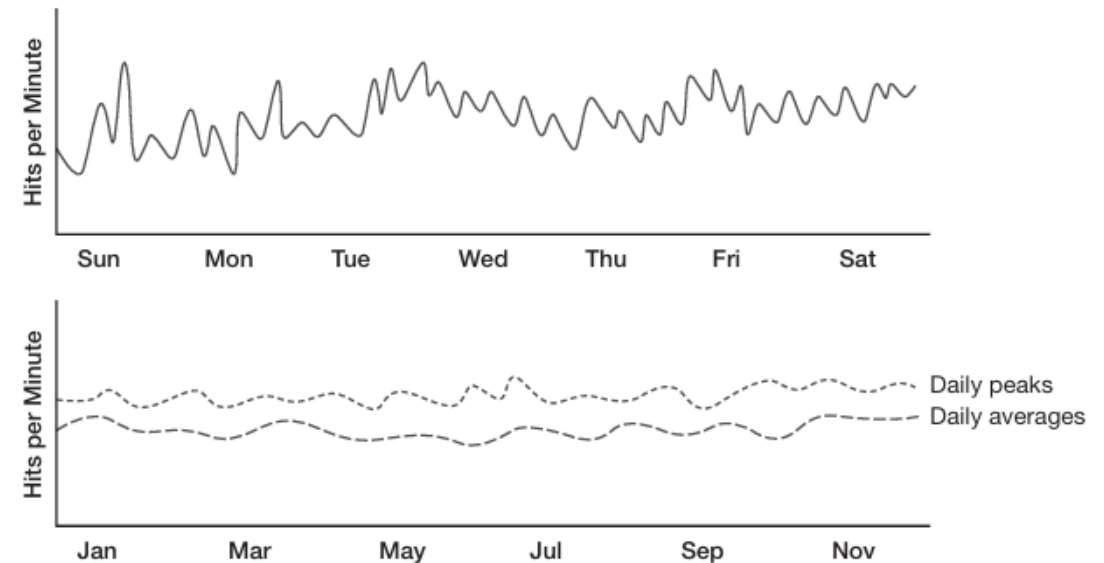
- **Step 3:** Know the Business Value of the System
 - Understand how important the system is to the business.
 - Decide when adding more capacity is useful.
 - Avoid adding capacity when it does not provide business value.
- Capacity planning requires understanding demand, evaluating current system performance, and deciding whether extra capacity is worth the cost.

Determining Expected Demand

- Capacity planners must identify the resource demand of each application in the system. Individual application demands are then combined to understand the overall system workload.
- Demand patterns usually change over time, so planners need to analyze:
 - Hourly traffic
 - Daily traffic
 - Weekly traffic
 - Yearly traffic
- Traffic graphs help reveal workload patterns, including:
 - Peak usage periods
 - Low-demand periods
 - Repeated spikes and dips

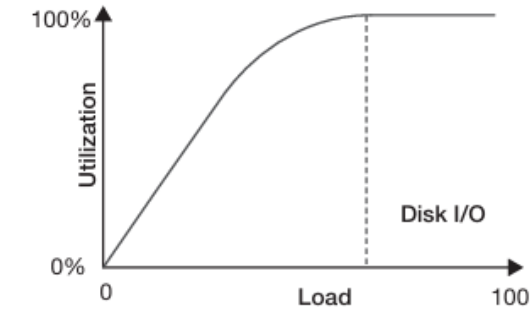
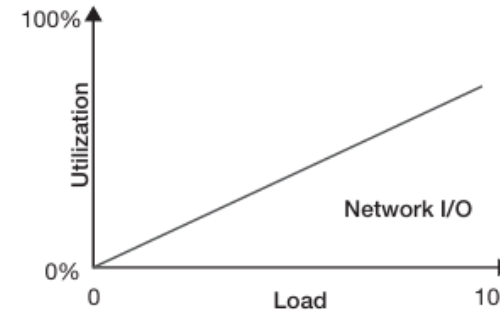
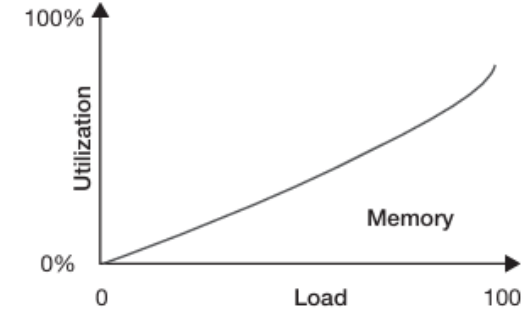
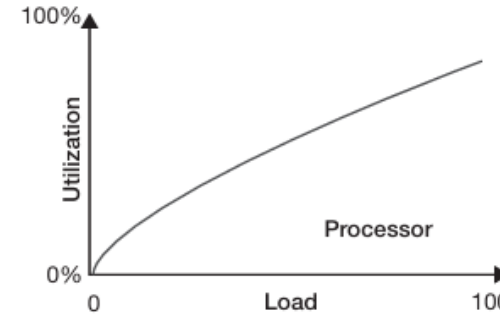
Determining Expected Demand

- Capacity planners should connect demand changes with business events, such as:
 - Product launches
 - Marketing campaigns
 - Holiday seasons
 - Annual festivals
- For new applications, planners may rely on business estimates until real usage data is available.
- Unexpected demand peaks cannot always be prevented, but they must be recognized and handled properly.



Analyzing Current Load

- System load is analyzed by measuring the usage of major computing resources:
 - Processor or CPU
 - Memory
 - Storage or disk I/O
 - Network connectivity
- Each resource has a utilization rate that increases as system load grows.
- When a resource reaches its maximum capacity, it creates a **stress point**.
- A stress point limits overall system performance and may cause the system to slow down or fail.



Analyzing Current Load

- Capacity planners use load testing to identify which resource becomes the bottleneck first.
- The first bottleneck is called the initial **stress point**.
- Adding more servers or resources may not always solve the problem if the real bottleneck is not identified.
- Resource utilization curves help show how each resource behaves under increasing load.
- In the example, disk I/O reaches its ceiling first at around 60% load, making it the initial stress point.
- After fixing the first bottleneck, load testing should be repeated to find the next performance limit.

Review

- Is the traditional N+1 rule of capacity planning applicable in cloud computing?
- Can a cloud service consumer avoid capacity planning through auto-scaling?
- What role does the service level agreement play in capacity planning?
- How do the IaaS provider manage capacity planning?
- Is it always clever to add capacity to support additional application demand?